

PROJECT PRESENTATION

Student Learning Management System

A full-stack architecture built using MongoDB, Express, React, and Node.js for modern educational scaling.



PROJECT OVERVIEW

This project aims to bridge the gap between traditional learning and digital accessibility by providing a centralized platform for students.

The Core Challenge

Dispersed learning materials and lack of structured progress tracking hinder academic growth.

Our Solution

A responsive MERN platform allowing students to browse courses, watch video content, and track completion through a unified dashboard.

Digital Learning



THE MERN TECH STACK

 MERN Icons

- ✓ **MongoDB:** Flexible NoSQL document-based database for course and user schemas.
- ✓ **Express.js:** Lightweight server-side framework for building robust REST APIs.
- ✓ **React:** Component-based frontend for a dynamic, high-performance user experience.
- ✓ **Node.js:** Scalable JavaScript runtime powering the backend application tier.

CORE STUDENT FEATURES



Course Discovery

Advanced filtering and search functionality to find relevant subjects instantly.



Video Learning

Integrated video player with progress persistence using YouTube/Vimeo APIs.



Progress Analytics

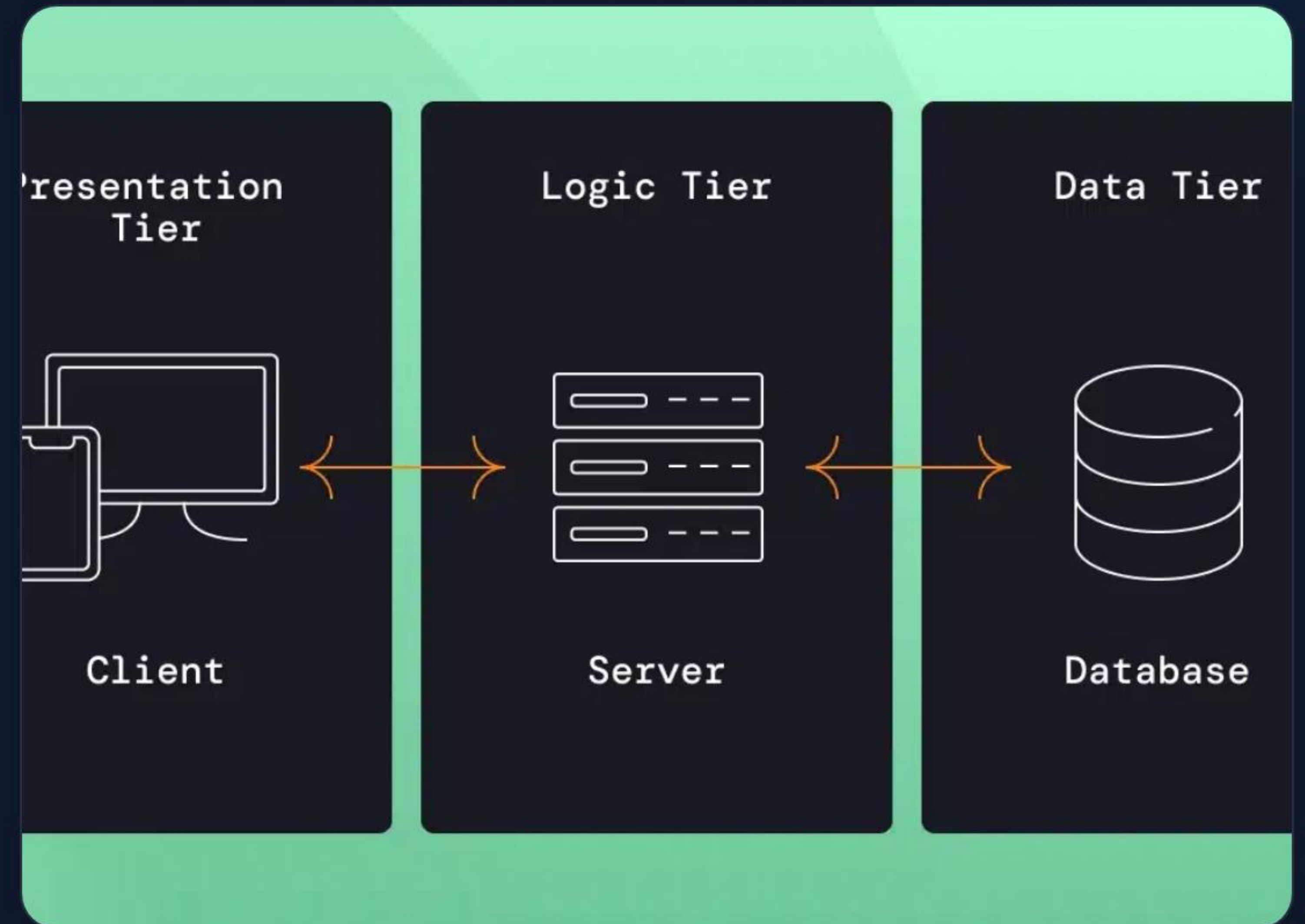
Visual tracking of course completion percentages and milestones achieved.

| SYSTEM ARCHITECTURE

3-Tier Design

The system follows a standard decoupling of layers for maximum scalability:

1. **Presentation Tier:** React SPA communicating via Axios.
2. **Application Tier:** Express controllers handling business logic.
3. **Data Tier:** MongoDB Atlas for cloud-native storage.



DATABASE SCHEMA DESIGN

Collection	Key Fields	Relationship
Users	Name, Email, Hash, Role (Student)	1-to-Many with Enrollments
Courses	Title, Description, Instructor, Price	1-to-Many with Lessons
Lessons	VideoURL, Duration, OrderIndex	Belongs to Course
Progress	UserID, CourseID, CompletedLessons	Pivot for Analytics

| SECURITY & LOGIC

- ✓ **JWT Auth:** Stateless authentication using JSON Web Tokens.
- ✓ **Bcrypt Hashing:** 12-round salt hashing for user password storage.
- ✓ **Middleware:** Role-based access control (RBAC) on sensitive routes.
- ✓ **CORS Protection:** Restricting API access to approved frontend domains.

Core Auth Logic

```
const token = jwt.sign({id: user._id}, secret, {  
  expiresIn: '24h' });  
  
res.cookie('token', token, { httpOnly: true });
```

STATE MANAGEMENT

Handling user sessions and course data across the Single Page Application (SPA) using React Context API or Redux Toolkit.



Auth Context

Global storage for current user profile and tokens.



Course Store

Caching fetched courses to reduce redundant API calls.



Persistence

Syncing state with LocalStorage for session recovery.

BACKEND API ARCHITECTURE

Student Routes

GET /api/courses/all

POST /api/courses/enroll/:id

GET /api/student/dashboard

PUT /api/student/update-progress

Design Principles

- ✓ Standard HTTP status codes (200, 201, 401, 500)
- ✓ JSON payload structure for every response
- ✓ Input validation using 'express-validator'

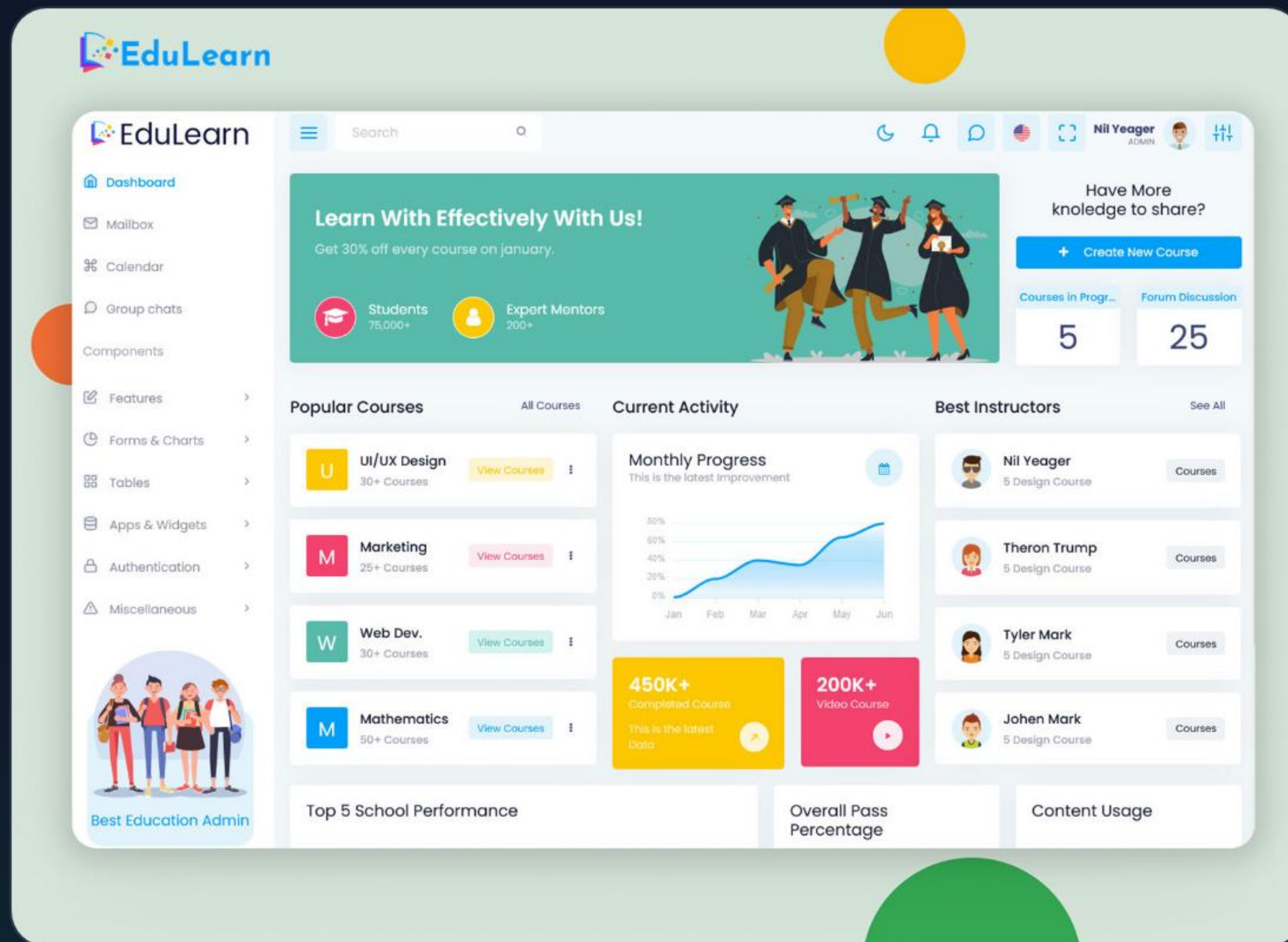
PROGRESS CALCULATION LOGIC

Tracking student completion is calculated dynamically on the server-side to ensure integrity:

$$P = \frac{L_{\text{completed}}}{L_{\text{total}}} \times 100$$

Where **P** is the Progress Percentage, **L-completed** is the count of unique lessons finished, and **L-total** is the total course content count.

LMS DASHBOARD VISUALIZATION



Modern Styling

Built with **CSS Modules** for component isolation and responsive layouts.

React Frontend

95% Mobile Ready

Node Backend

85% API Uptime

Digital Learning



Future Enhancements

Incorporating AI-driven course recommendations and real-time Peer-to-Peer messaging for enhanced student engagement.

Any Questions?

Full source code available on [GitHub Repo](#)

IMAGE SOURCES



https://static.vecteezy.com/system/resources/previews/045/374/711/non_2x/digital-learning-concept-a-student-in-graduation-cap-multitasks-with-a-laptop-and-a-book-symbolizing-modern-education-and-information-access-illustration-vector.jpg

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